

Path & Query Parameters in FastAPI

This is one of the **most important concepts in APIs**, because it controls how users send data to your server.

You'll learn:

- What Path Parameters are
- What Query Parameters are
- How FastAPI uses them
- Differences between them
- Status codes (very important in APIs)
- Code examples + real-world understanding

1. Path Parameters

What is a Path Parameter?

A **Path Parameter** is a value that is part of the URL itself.
It is used to identify a specific resource.

Simple Definition:

Path parameter = data inside the URL path

Example URL:

`/students/5`

Here:

- students = collection
- 5 = specific student ID (path parameter)

Real-life analogy:

Imagine a library:

- /book/10 → means “give me book number 10”

You are directly pointing to a specific item.

FastAPI Example:

```
from fastapi import FastAPI

app = FastAPI()
```

```
@app.get("/students/{student_id}")
def get_student(student_id: int):
    return {"student_id": student_id}
```

What happens?

If you open:

```
/students/7
```

Response:

```
{
  "student_id": 7
}
```

Key Idea:

Path parameters are **required values**.

If missing → request fails.

2. Code Demo (Path Params)

Typical demo includes:

Example with multiple parameters:

```
@app.get("/students/{student_id}/courses/{course_id}")
def get_data(student_id: int, course_id: int):
    return {
        "student_id": student_id,
        "course_id": course_id
    }
```

URL Example:

```
/students/3/courses/10
```

Output:

```
{
  "student_id": 3,
  "course_id": 10
}
```

3. Path Function (FastAPI Features)

FastAPI gives special tools for path parameters.

1. Type Validation

```
@app.get("/items/{item_id}")
def read_item(item_id: int):
    return {"item_id": item_id}
```

FastAPI ensures:

- Only integers are allowed
- Wrong input → error automatically

2. Data Conversion

If you pass:

```
/items/5
```

FastAPI converts "5" → int(5)

3. Built-in Error Handling

If invalid:

```
/items/abc
```

Response:

- Error automatically generated (no extra code needed)

4. Path Converters (Important)

You can control data type:

Type	Meaning
int	number
str	text
float	decimal
bool	true/false

3. HTTP Status Codes

Status codes tell us:

What happened with the request

Simple Meaning:

Status code = server response message

Common Status Codes:

Code	Meaning	When used
200	OK	Success
201	Created	New data added
400	Bad Request	Wrong input
404	Not Found	Resource missing
500	Server Error	Backend failure

Example:

If student not found:

```
{  
  "detail": "Not Found"  
}
```

Status:

404

Why important?

Because APIs are not just data:

👉 They also communicate **status of operations**

4. Query Parameters

Now the second major concept.

What is a Query Parameter?

Query parameters are:

Extra data added after ? in URL

Example:

```
/students?age=20
```

Here:

- age=20 is query parameter

Simple Definition:

Query parameter = optional filters or settings in URL

Real-life analogy:

Imagine online shopping:

- /products?price=low

- /products?brand=nike

You are filtering results.

FastAPI Example:

```
@app.get("/students")
def get_students(age: int = None):
    return {"age_filter": age}
```

Request:

```
/students?age=25
```

Response:

```
{
  "age_filter": 25
}
```

Key Idea:

Query parameters are **optional**

If not given:

- Default value is used

Path vs Query Parameters (VERY IMPORTANT)

Feature	Path Param	Query Param
Location	URL path	After ?
Required	Yes	No
Purpose	Identify resource	Filter or modify
Example	/students/5	/students?age=20

5. Code Demo (Query Params)

Combined example:

```
@app.get("/students/{student_id}")
def student_details(student_id: int, grade: str = None):
    return {
        "student_id": student_id,
        "grade": grade
    }
```

Request:

```
/students/5?grade=A
```

Response:

```
{  
  "student_id": 5,  
  "grade": "A"  
}
```

Summary**Path Parameters:**

- Inside URL
- Required
- Identify specific resource

Example:

```
/students/10
```

Query Parameters:

- After ?
- Optional
- Used for filtering/searching

Example:

```
/students?age=20
```

HTTP Status Codes:

- Tell result of request
- Important for debugging APIs

Real-world Understanding**Instagram example:**

- /user/123 → specific user (Path)
- /posts?likes=100 → filter posts (Query)

Why this matters (ML + APIs)

In ML APIs:

- Path → model selection (/model/1)

- Query → parameters (?threshold=0.8)
-

FastAPI uses:

- Path params → structure of data
- Query params → flexibility in requests
- Status codes → communication of results